DXC Technology & International University of Rabat (UIR)

FX Trading Room P&L; Engine — Defense Executive Guide

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Project Scope: Institutional FX Trading Room Position Keeping & Mark-to-Market Engine	**Primary Fixing:** Bank Al-Maghrib (BAM) Official Fixing (15:30)
Core Accounting: Multi-Currency P&L; in MAD (FIFO vs. CMP/WAC)	**Storage Architecture:** PostgreSQL 16 (Docker) + SQLite 3 Resilient Fallback
Pipeline Orchestration: Apache Airflow 2.9.3 + dbt Data Transformation	**Verification Status:** 10/10 PyTest Passed | 6/6 dbt Tests Passed

1. Executive Summary & The Core Problem

A corporate treasury or commercial bank operating in Morocco cannot manage foreign exchange risk across disparate, non-standardized systems. Trading desks execute transactions in multiple global currencies (USD, EUR, GBP), but Moroccan accounting standards require every single position, realized gain, and balance sheet provision to be reported strictly in **Moroccan Dirham (MAD)** based on Bank Al-Maghrib central fixings.

The 3 Fundamental Operational Challenges Solved:

- **Currency Conversion & Peg Compliance (BR1 & BR2):** Real-time conversion of direct MAD pairs and synthetic cross-rates (EUR/USD) into MAD, with daily MTM revaluation against BAM's 60/40 currency basket.
- **Inventory Depletion Ambiguity (BR4):** FIFO (First-In, First-Out) recognizes trading gains earlier, whereas CMP (Weighted Average Cost) smooths volatile exchange rate spikes across reporting periods.
- **Front-to-Back Office Discrepancies (BR5):** Automatic detection of rate slippages, partial nominal fills, unconfirmed SWIFT MT300 affirmations, and orphan back-office settlement tickets.
- **Mathematical Accounting Balance (BR3):** Strictly enforces and mathematically verifies that Total P&L; = Realized P&L; + Unrealized P&L; with zero tolerance for discrepancies.

2. The Medallion Data Engineering Pipeline

Our data architecture follows the industry-standard Medallion model, orchestrated end-to-end by Apache Airflow:

Pipeline Stage	Technology	Role & Transformations Performed
1. Raw Ingestion	Python / REST / CSV	Pulls trades from Front-Office blotters and official Bank Al-Maghrib daily fixings.
2. Bronze Staging	PostgreSQL / SQLite	Raw immutable persistence into stg_trade and stg_market_rate.
3. Silver SQL Marts	dbt (Data Build Tool)	SQL transformations building fact_trade_mtm: computes synthetic EUR/USD cross into MAD.
4. Gold P&L; Engine	Python (deque / math)	Chronological FIFO queue lot matching and WAC rolling cost calculations.
5. Persistence & Mart	PostgreSQL / SQLite	Upserts into daily_pnl with physical database CHECK constraints.
6. Quality Assertions	dbt test & PyTest	6 automated dbt tests + 10 unit tests halting the pipeline upon any anomaly.

3. Step-by-Step Running Guide (Mac & Windows)

You can run the application in two modes: **Option A (Instant Zero-Docker)** or **Option B (Full Enterprise Stack)**.

Option A: Instant Zero-Docker Run (Fastest for Defense / Laptops)

Uses the local SQLite database (db/fx_trading_room.db) with all 72 Moroccan trades already populated. Requires zero Docker containers.

Terminal Commands (Mac / Windows):

```
git pull origin main
pip install -r requirements.txt
streamlit run dashboard/app.py
```

*Result: Dashboard opens instantly at **http://localhost:8501** with all trades, charts, and blotters.*

Option B: Full Enterprise Stack (PostgreSQL 16 + Airflow + dbt)

Runs the complete containerized architecture with PostgreSQL on port 5432 and Apache Airflow on port 8080.

Terminal Commands:

```
# 1. Start Docker containers in background
docker compose up -d
# 2. Access Apache Airflow Web UI
Open http://localhost:8080 (Username: admin | Password: admin)
# 3. Launch Streamlit (automatically detects live PostgreSQL)
streamlit run dashboard/app.py
```

*Result: Sidebar displays **'Storage Engine: PostgreSQL 16 (Live Docker Container)'**.*

4. Live Terminal Verification Commands (For the Jury)

If a jury member asks to see live tests in the terminal, run these exact commands:

Audit Objective	Command to Run in Terminal	Expected Clean Output
dbt Data Quality Tests (6 Pipeline Checks)	`docker exec fx-pnl-airflow bash -c "cd /opt/airflow/fx_pnl_dbt && dbt test"` *(or `cd fx_pnl_dbt && dbt test`)*	**Done. PASS=6 WARN=0 ERROR=0** Tests BR3 identity, positive nominals, positive rates, unique IDs.
PyTest Full Test Suite (10 Python Unit Tests)	`pytest -v`	**10 passed in 0.55s (100%)** Validates MTM math, FIFO deque, CMP cost, and FO/BO matching.
SQL Integrity Check (Zero Accounting Breaks)	`sqlite3 db/fx_trading_room.db "SELECT * FROM daily_pnl WHERE ROUND(total_mad, 2) != ROUND(realized_mad + unrealized_mad, 2);"`	**(0 rows)** Mathematically proves zero balance sheet discrepancies exist.

5. Interactive UI Demo Tour (What to Click & Show)

Demonstrate the platform's professional capabilities following this exact sequence:

UI Component	Action / Click	Key Financial & Technical Concept to Highlight
Sidebar: Storage Engine Switcher	Switch between *PostgreSQL* and *SQLite*	Proves **resilient dual-engine architecture**: zero crash fallback when offline, enterprise containerization when online.
Trading Period Calendar Filter	Select *Filter by Month* or *Specific Day*	Shows dynamic KPI recalculation (Total MTM, Realized, Latent) across 72 institutional trades (August to October 2026).
PFE Benchmark Button	Select *PFE Benchmark (2026-09-07)*	Reproduces the exact brief benchmark: **+79,460 MAD Total MTM** (+20,000 Realized, +59,460 Latent).
Tab 1: Active Trade Blotter	Show formulas column	Full mathematical transparency: displays exact arithmetic formula in MAD for every trade (e.g. (close - trade) * nominal).
Tab 2: FO/BO Reconciliation Hub	Filter by *Breaks Only*	Detects **4 authentic banking breaks**: rate slippage (+3.5 pips), partial nominal cut, unconfirmed SWIFT MT300, and orphan settlement.
Tab 3: BAM Market Fixings Terminal	Toggle USD/MAD, EUR/MAD, EUR/USD	Pure institutional FX candlestick charting with 20-period Simple Moving Average and central bank fixing indicators.
Tab 4: FIFO vs. CMP Inventory	Review Section 4 proof	Demonstrates lot unwinding: FIFO Realized = **+22,000 MAD** vs. CMP Realized = **+18,000 MAD** (explains the brief typo).
Tab 5: BAM Fluctuation Band	View +/-5% Stress Test Matrix	Evaluates desk stop-loss compliance against Bank Al-Maghrib peg boundaries (60% EUR / 40% USD basket).
Tab 6: Airflow & Data Engineering	Click *Trigger Airflow DAG*	Executes the full pipeline live and provides direct hyperlink to Airflow Web UI on port 8080.

6. Anticipated Jury Questions & Winning Answers

Q1: Why is FIFO realized profit (+22,000 MAD) higher than CMP (+18,000 MAD)?

Answer: 'In a rising exchange rate market, FIFO depletes the oldest, cheaper lots first (purchased at 9.9100), yielding a higher immediate realized gain on sale (+22k MAD). CMP averages all purchase prices (9.9200), yielding +18k MAD. This is purely a timing difference: upon complete liquidation of the book, cumulative P&L; converges to identical values.'

Q2: How do you value currency pairs without a direct BAM fixing (e.g. EUR/USD)?

Answer: 'We construct a synthetic cross-rate via triangular arbitrage: EUR/USD = (EUR/MAD) / (USD/MAD). This guarantees consistent Mark-to-Market revaluation in MAD without introducing cross-rate arbitrage breaks.'

Q3: Why test data quality at both the database level (CHECK) and the pipeline level (dbt)?

Answer: 'We apply Defense-in-Depth: database CHECK constraints prevent physically corrupt rows from being stored, while dbt tests act as automated quality gates in Airflow. If an upstream market feed sends invalid data, dbt detects it and halts the pipeline before incorrect figures reach risk managers.'